

SLIDRONIX

Phase F → Phase G

Deep Learning, Multimodal Fusion & Model Comparison

Detailed Technical Reference Document

Prepared as a single study/review reference for implementation, viva, documentation and presentation.

Scope of this document: Phase F and Phase G only. Phase H is intentionally excluded.

1. Where Phase F and Phase G Fit

The completed pipeline entering Phase F contains a balanced binary landslide dataset with terrain, satellite, soil and rainfall information. Phase F converts those inputs into several learning architectures. Phase G then compares those architectures using a common evaluation framework and adds a classical Random Forest benchmark.

Stage	Purpose	Main output
Phase A	Historical landslide inventory, preprocessing and baseline model construction	Overlaid historical classification baseline RF
Phase B	Terrain features from DEM	Elevation and slope
Phase C	Satellite vegetation/land-cover + soil	NDVI, land cover and six soil properties
Phase D	Rainfall temporal features	30-day daily sequence + rainfall summaries
Phase E	Binary dataset construction and spatial leakage control	Balanced train/validation/test datasets
Phase F	Deep-learning and multimodal learning	MLP, CNN, Transformer and Fusion models
Phase G	Model comparison and classical benchmark	RF + common comparison table/plots

2. Phase E Inputs Used by Phase F

Phase E produced 13,982 balanced samples: 6,991 background and 6,991 landslide. The spatial split was fixed before model training: 8,385 training rows, 2,797 validation rows and 2,800 test rows. Duplicate-coordinate leakage was controlled using spatial blocks.

Important: the binary target is Landslide = 1 and Background = 0. This is different from the Phase A Random Forest, whose target was historical *Movement Type*. Phase G therefore trains a new binary RF for a fair comparison with the Phase F binary models.

Split	Rows	Background	Landslide	Spatial blocks
Train	8,385	4,192	4,193	881
Validation	2,797	1,398	1,399	291
Test	2,800	1,401	1,399	301
Total	13,982	6,991	6,991	1,473 unique blocks

3. Phase F — Objective

Phase F is the deep-learning stage. Instead of depending on a single model type, the project represents the same landslide problem through three complementary modalities: tabular environmental conditions, spatial terrain structure, and temporal rainfall behaviour. A fourth model fuses these representations.

Core idea: different model families are given the kind of data they are designed to represent. The MLP learns relationships among numerical/categorical environmental features; the CNN learns local spatial terrain patterns; the Transformer learns relationships across the 30 daily rainfall observations; and the Fusion model combines all three learned representations.

Final prediction: a single logit is converted with a sigmoid function into a probability between 0 and 1. This probability is used as the model's landslide-risk score in the later Phase H pipeline.

4. Phase F Input Modalities

Modality	Input	Representation	Why this representation is used
Tabular	26 environmental features	Vector of 26 values	Suitable for structured heterogeneous features
Spatial	Elevation + slope	$2 \times 32 \times 32$ terrain patch	Preserves local neighbourhood structure
Temporal	30 daily rainfall values	30×1 sequence	Preserves order across rainfall days
Multimodal	All three above	Three learned embeddings	Combines complementary information

5. The 26 Tabular Features

No.	Feature	Role / interpretation
1	Elevation_m	Terrain elevation.
2	Slope_deg	Terrain steepness in degrees.
3	rainfall_total_30d_mm	Rainfall-derived summary feature.
4	rainfall_mean_daily_mm	Rainfall-derived summary feature.
5	rainfall_max_daily_mm	Rainfall-derived summary feature.
6	rainfall_std_daily_mm	Rainfall-derived summary feature.
7	rainfall_7d_mm	Rainfall-derived summary feature.
8	rainfall_15d_mm	Rainfall-derived summary feature.
9	rainfall_30d_mm	Rainfall-derived summary feature.
10	rainfall_max_7d_mm	Rainfall-derived summary feature.
11	rainfall_max_15d_mm	Rainfall-derived summary feature.
12	rainfall_max_30d_mm	Rainfall-derived summary feature.
13	NDVI_mean	Mean vegetation index over valid 2022 observations.
14	NDVI_std	Temporal variability of NDVI.
15	NDVI_min	Minimum valid NDVI.
16	NDVI_max	Maximum valid NDVI.
17	NDVI_valid_observations	Count of valid NDVI observations.
18	NDVI_range	NDVI max – min.
19	Land_Cover_Type	MCD12Q1 land-cover class.
20	Land_Cover_QC	Land-cover quality/control field.
21	Soil_clay_0_5cm	Clay content, 0–5 cm.
22	Soil_sand_0_5cm	Sand content, 0–5 cm.
23	Soil_silt_0_5cm	Silt content, 0–5 cm.
24	Soil_soc_0_5cm	Soil organic carbon.
25	Soil_bdod_0_5cm	Bulk density.
26	Soil_phh2o_0_5cm	Soil pH.

6. Preprocessing and Data Leakage Control

The training scripts fit preprocessing statistics using the training data and then apply those transformations to validation and test data. This is important because calculating scaling or imputation statistics from the test set would leak information from the evaluation set into the model.

6.1 Tabular preprocessing

The tabular branch uses median imputation for missing numerical values followed by StandardScaler. The training split determines the imputation/scaling parameters.

NDVI contains a small number of missing values because of quality filtering. The model therefore does not require artificial replacement values to be inserted into the original scientific dataset; missing values are handled in the modelling pipeline.

6.2 CNN preprocessing

The elevation and slope channels are standardized using channel statistics calculated from the CNN training patches. The same training-derived statistics are applied to validation and test patches.

6.3 Rainfall preprocessing

The 30 daily rainfall sequence is standardized using training-set mean and standard deviation. The same transformation is applied to validation and test sequences.

Rainfall limitation: the current sequence is a fixed January 1–30, 2022 reference period. It is not an event-aligned sequence immediately preceding each landslide. Therefore, the Transformer result should be interpreted as temporal-pattern modelling for the available reference period, not as proof of real-time trigger prediction.

7. Phase F Model 1 — MLP

The Multilayer Perceptron is the structured-data model. It receives the 26 tabular features and learns nonlinear combinations among terrain, rainfall summaries, vegetation, land cover and soil variables.

Layer	Configuration	Purpose
Input	26 features	Receives standardized/imputed tabular data
Hidden 1	128 units + BatchNorm + ReLU + Dropout 0.30	Initial nonlinear feature transformation
Hidden 2	64 units + ReLU + Dropout 0.25	Higher-level representation
Hidden 3	32 units + ReLU	Compact embedding
Output	1 unit	Binary landslide logit
Loss	BCEWithLogitsLoss	Binary classification objective
Optimizer	Adam, lr 0.001, weight decay 1e-4	Parameter optimization
Training	12 epochs, batch 64	Iterative learning

Why MLP? Structured environmental data does not inherently form an image or a sequence. An MLP is therefore a direct neural baseline for the 26-feature vector.

8. Phase F Model 2 — CNN

The CNN is the spatial model. Each sample is represented by a 32 × 32 neighbourhood around the point, with two channels: elevation and slope. This allows the network to learn local terrain texture and spatial gradients rather than seeing only the single elevation/slope values at the centre point.

Component	Configuration
Input	$2 \times 32 \times 32$
Block 1	Conv $2 \rightarrow 32$, BatchNorm, ReLU, MaxPool
Block 2	Conv $32 \rightarrow 64$, BatchNorm, ReLU, MaxPool
Block 3	Conv $64 \rightarrow 128$, BatchNorm, ReLU, AdaptiveAvgPool
Projection	$128 \rightarrow 64 \rightarrow 1$
Loss	BCEWithLogitsLoss
Optimizer	Adam, lr 0.001
Training	8 epochs, batch 64

The spatial dataset contained valid patches for 8,224/8,385 training samples, 2,726/2,797 validation samples and 2,761/2,800 test samples. The missing patches were excluded from CNN evaluation. This is why CNN and Fusion test sample counts are 2,761 rather than 2,800.

9. Phase F Model 3 — Rainfall Transformer

The Transformer receives the actual 30 daily rainfall values, rather than only aggregate rainfall statistics. Each day is represented as one timestep with one rainfall value.

Component	Configuration
Sequence	30 days $\times$ 1 rainfall value
Projection	$1 \rightarrow 64$
Position encoding	Learnable 30-position embedding
Transformer	2 encoder layers, 4 attention heads, feed-forward 128, dropout 0.2
Pooling	Mean across 30 encoded timesteps
Projection	$64 \rightarrow 32$
Output	1 binary logit
Training	Adam lr 0.001, 12 epochs, batch 64

What attention does here: the model can learn relationships between different positions in the rainfall sequence. It is not simply calculating a total; it learns a representation of the 30-day pattern.

10. Phase F Model 4 — Multimodal Fusion

The Fusion model is the central multimodal architecture. It does not put all raw features into one giant vector. Instead, each modality first passes through its own encoder. The resulting embeddings are then concatenated and passed through a fusion network.

Branch	Output embedding	Main encoder
Tabular	32	$26 \rightarrow 128 \rightarrow 64 \rightarrow 32$
Spatial CNN	32	$2 \times 32 \times 32 \rightarrow$ convolution blocks $\rightarrow 128 \rightarrow 32$
Rainfall Transformer	32	$30 \times 1 \rightarrow 64 \rightarrow$ Transformer $\rightarrow 64 \rightarrow 32$
Combined	96	$32 + 32 + 32$
Fusion head	1 logit	$96 \rightarrow 64 \rightarrow 32 \rightarrow 1$

The exact Fusion implementation uses a 96-dimensional concatenated representation. The fusion head applies a 96→64 transformation with BatchNorm, ReLU and dropout, followed by 64→32, ReLU, dropout and a final 32→1 output.

The trained Fusion checkpoint is loaded for evaluation using the same architecture and state-dictionary structure used during training. This is important for reproducibility: inference must reconstruct the same layer names and dimensions as training.

11. Fusion Forward Pass — Step by Step

- Step 1 — Tabular encoder: the 26-feature vector becomes a 32-dimensional tabular embedding.
- Step 2 — CNN encoder: the 2-channel terrain patch is passed through convolutional layers and spatial pooling, then projected to 32 dimensions.
- Step 3 — Rainfall projection: each daily rainfall value is projected from 1 dimension to 64 dimensions.
- Step 4 — Position information: a learnable 30-position embedding is added so the model can distinguish different days.
- Step 5 — Transformer encoder: the 30 rainfall tokens interact through self-attention and feed-forward layers.
- Step 6 — Temporal pooling: the 30 encoded rainfall representations are averaged into one 64-dimensional representation, then projected to 32 dimensions.
- Step 7 — Fusion: the three 32-dimensional embeddings are concatenated into 96 dimensions.
- Step 8 — Classification head: the 96-dimensional vector is transformed through dense layers to one logit.
- Step 9 — Probability: sigmoid(logit) produces a value from 0 to 1.

12. Phase F Training Concept — What Happens in an Epoch?

An **epoch** means one complete pass through the training dataset. For example, with a batch size of 64, the model does not process all 8,224 CNN training samples at once. It processes them in batches, calculates the loss, performs backpropagation, updates weights, and continues until the training set has been traversed.

A simplified training loop is:

```
Input batch → forward pass → predicted logit → loss → backpropagation → gradients → optimizer update → next batch.
```

After an epoch, validation data is evaluated without updating weights. Validation AUC is used to identify the best checkpoint in the Phase F scripts. The test set is kept for final evaluation.

13. Phase F Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Test samples
MLP	0.8236	0.7891	0.8828	0.8333	0.8968	2,800
CNN	0.8124	0.7747	0.8831	0.8254	0.8815	2,761
Transformer	0.6854	0.7144	0.6169	0.6621	0.7322	2,800
Fusion	0.8218	0.8033	0.8543	0.8280	0.9122	2,761

These are test-set measurements from the current implementation. They describe the behaviour of each model on the held-out test data under the project's present dataset construction and preprocessing.

Metric-specific observations: the Fusion model has the highest ROC-AUC among the listed Phase F models at 0.9122. The MLP has accuracy 0.8236 and F1 0.8333. The CNN has recall 0.8831. The Transformer has ROC-AUC 0.7322. Because CNN/Fusion use 2,761 valid spatial samples, their test coverage differs from the 2,800-sample MLP/Transformer evaluation.

14. Understanding the Metrics

Metric	Meaning
Accuracy	Fraction of all test samples classified correctly.
Precision	Among samples predicted as landslide, fraction that are actually landslide.
Recall	Among actual landslide samples, fraction detected by the model.
F1	Harmonic mean of precision and recall.
ROC-AUC	Measures ranking/separation ability across classification thresholds; 0.5 corresponds to random ranking.

For an early-warning-oriented system, recall is important because missed landslide cases matter. Precision is also important because excessive false alarms reduce the usefulness of an alerting system. ROC-AUC is useful because it evaluates ranking across thresholds rather than only one fixed threshold.

15. Phase F Confusion Matrices

Model	TN	FP	FN	TP
MLP	1,071	330	164	1,235
CNN	1,019	356	162	1,224
Transformer	1,056	345	536	863
Fusion	1,085	290	202	1,184

TN = correctly identified background; FP = background predicted as landslide; FN = landslide predicted as background; TP = correctly identified landslide. These values explain the metric differences. For example, the Transformer has substantially more false negatives than the other models, which corresponds to its lower recall.

16. Phase F Files

File	Purpose
src/train_mlp.py	Trains the tabular MLP and saves its best checkpoint/results.
src/test_cnn_patch.py	Checks that terrain patches can be extracted around a sample point.
src/create_cnn_spatial_dataset.py	Builds elevation+slope 32×32 CNN datasets.
src/train_cnn.py	Trains the spatial CNN.
src/train_transformer.py	Trains the 30-day rainfall Transformer.
src/train_fusion.py	Builds and trains the exact multimodal Fusion model.
outputs/phase_f/*.pt	Saved PyTorch model checkpoints.
outputs/phase_f/*_results.txt	Training/test result records.

17. Phase G — Why a Separate Comparison Phase?

Phase G answers a different question from Phase F. Phase F demonstrates and trains multiple architectures. Phase G establishes a structured comparison under the same project dataset and evaluation split, and adds a classical machine-learning benchmark.

The old Phase A Random Forest cannot simply be copied into the Phase G table because its target was Movement Type. Phase G instead trains a new binary Random Forest using the same 26-feature environmental representation and the Phase E binary target.

18. Phase G Random Forest

Parameter	Value
Algorithm	RandomForestClassifier
Trees	300
Maximum depth	25
Minimum samples per leaf	2
Max features	sqrt
Class weighting	balanced
Random state	42
Parallelism	n_jobs=-1
Preprocessing	Median imputation
Features	26
Target	Landslide (0/1)

The RF receives the structured 26-feature table. It does not use the CNN terrain patch and does not use the Transformer sequence as a separate modality. This makes it a classical tabular benchmark against the neural tabular and multimodal approaches.

19. Phase G Random Forest Results

Split	Accuracy	Precision	Recall	F1	ROC-AUC
Validation	0.8023	0.8195	0.7756	0.7969	0.8939
Test	0.8154	0.8286	0.7949	0.8114	0.9093

Test confusion matrix: TN = 1,171; FP = 230; FN = 287; TP = 1,112.

The Phase G RF is therefore a meaningful classical reference model. Its test ROC-AUC is 0.9093, while the current Fusion model reports 0.9122 on its valid spatial test subset. The difference should be interpreted with the sample-coverage caveat described below.

20. Random Forest Feature Importance

Rank in RF importance output	Feature	Importance
1	Slope_deg	0.203980
2	Soil_soc_0_5cm	0.090283

Rank in RF importance output	Feature	Importance
3	Soil_bdod_0_5cm	0.078753
4	Elevation_m	0.057053
5	NDVI_mean	0.052864
6	Land_Cover_Type	0.047607
7	Soil_clay_0_5cm	0.039483
8	Soil_sand_0_5cm	0.038037
9	Soil_phh2o_0_5cm	0.035650
10	rainfall_std_daily_mm	0.032805

These are model-derived Random Forest feature-importance values. They indicate which features contributed most to the RF's split-based decision process. They should not automatically be interpreted as causal effects or as proof that a feature independently causes landslides.

21. Phase G Comparison Table

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Test samples
Random Forest	0.8154	0.8286	0.7949	0.8114	0.9093	2,800
MLP	0.8236	0.7891	0.8828	0.8333	0.8968	2,800
CNN	0.8124	0.7747	0.8831	0.8254	0.8815	2,761
Transformer	0.6854	0.7144	0.6169	0.6621	0.7322	2,800
Fusion	0.8218	0.8033	0.8543	0.8280	0.9122	2,761

The table is a metric-by-metric comparison, not a single overall ranking. Different metrics emphasize different properties. Accuracy and F1, for example, reflect a fixed classification threshold, whereas ROC-AUC evaluates discrimination over thresholds.

22. How to Explain the Results in a Viva

Q: Why did you use multiple models?

A: The data has different structures. Environmental values are tabular, terrain has spatial structure, and rainfall is a time sequence. We therefore evaluated an MLP, CNN and Transformer separately, then built a Fusion model to combine their learned representations.

Q: Why is the Transformer lower?

A: The Transformer in this implementation receives only the fixed 30-day rainfall sequence, while the MLP receives 26 integrated environmental features. The current rainfall period is also a fixed January 2022 reference period rather than an event-aligned pre-landslide sequence. Therefore the rainfall-only model does not have access to the other environmental information.

Q: Why use CNN?

A: A single elevation and slope value describe only the centre point. The CNN receives a 32×32 terrain neighbourhood, allowing it to learn local spatial patterns.

Q: Why use Fusion?

A: Each modality has a different structure. Separate encoders preserve those structures before combining them. The Fusion network then learns how the tabular, spatial and temporal embeddings interact.

Q: Why is Random Forest still needed?

A: It provides a classical machine-learning benchmark. It is trained on the same binary task and 26 structured features, so the project can compare a conventional ensemble method against neural approaches.

Q: What does ROC-AUC 0.9122 mean?

A: It means the current Fusion model has strong ranking/separation performance on the evaluated test samples across classification thresholds. It is not a statement that 91.22% of predictions are correct.

Q: Can you call this a real-time early-warning model?

A: Not yet. The current rainfall input is a fixed January 1–30, 2022 reference sequence. Event-aligned rainfall and live rainfall integration are future enhancements required for a true operational early-warning workflow.

23. Important Evaluation Caveats

1. **Test-set coverage:** MLP, Transformer and RF use all 2,800 Phase E test rows. CNN and Fusion use 2,761 rows with valid terrain patches. Their metrics therefore do not come from exactly identical test subsets.
2. **Rainfall temporal alignment:** the 30-day rainfall sequence is fixed to January 2022, not individually aligned to each landslide event.
3. **Binary background construction:** background samples were generated using the project's spatial sampling rules. They represent sampled non-landslide locations, not a complete inventory of every location where no landslide ever occurred.
4. **Risk thresholds:** the later risk categories are project-defined classification thresholds and are not automatically validated emergency thresholds.
5. **Correlation versus causation:** model feature importance is predictive information, not proof of physical causality.
6. **Generalization:** results describe this dataset, study region, feature sources, sampling strategy and split. External geographic validation would be needed to establish broader generalization.

24. Reproducibility — Commands

From the project root:

```
python src/train_mlp.py
python src/create_cnn_spatial_dataset.py
python src/train_cnn.py
python src/train_transformer.py
python src/train_fusion.py
python src/train_rf_phase_g.py
python src/create_phase_g_comparison.py
```

The scripts save checkpoints and result files under **outputs/phase_f** and **outputs/phase_g**. The Phase G comparison is generated from the recorded model results.

25. One-Page Mental Model

Phase E → balanced binary dataset + spatially separated train/validation/test.

Phase F → learn four views of the problem:

- MLP → environmental feature relationships
- CNN → local elevation/slope structure
- Transformer → 30-day rainfall sequence
- Fusion → combines tabular + spatial + temporal embeddings

Phase G → add binary Random Forest benchmark and compare all model outputs using Accuracy, Precision, Recall, F1 and ROC-AUC.

Key result set: Fusion ROC-AUC 0.9122; RF ROC-AUC 0.9093; MLP F1 0.8333; CNN recall 0.8831; Transformer ROC-AUC 0.7322. These are metric-specific results, not a single overall ranking.

Next stage concept: use the trained prediction pipeline to produce spatial risk outputs and then build the application/dashboard layer. That is outside the scope of this Phase F→G reference.

End of Phase F → G Detailed Reference